Camera calibration checkerboard -- 9 x 6 INNER CORNERS, 20.0 mm squares

board 200.0 x 140.0 mm (10 x 7 squares) | inner-corner span 160.0 x 100.0 mm | page 2 is the board

1. **PRINT SETTINGS. Scale 100% / 'Actual size'. NEVER 'fit to page' or 'shrink oversized pages'.**
   Landscape. Duplex OFF. MATTE plain paper -- gloss reflects the light you are calibrating under.

2. **VERIFY THE PRINTER on the bars below BEFORE you mount the board.**
   200.0 mm bar and 150.0 mm bar, measured with a steel rule. Out by >1 mm -> fix the dialog and reprint.

3. **MOUNT IT DEAD FLAT. Spray-mount page 2 to foamboard, MDF, or glass -- no bubbles, no bow, no curl.**
   After scale, FLATNESS is the dominant calibration error: a 2 mm bow across the board biases the distortion coefficients and the RMS will still look fine. Tape at the corners only is NOT enough.

4. **MEASURE THE BOARD ITSELF (the board is its own ruler -- do this even if you bought a board).**
   Outermost inner corner to outermost inner corner: 160.0 mm along the LONG axis, 100.0 mm along the short axis. Full black extent: 200.0 x 140.0 mm. Divide the long span by 8 to get your ACTUAL square size and pass THAT to --square. Do not assume the nominal value.

5. **CAPTURE >= 15 views (build_tab skr-06 acceptance: RMS reprojection <= 1.0 px).**
   Vary angle, distance and position; fill the corners of the frame, not just the centre; keep the board sharp (it is the LENS you are measuring, not the paper). Shoot in the FINAL tripod geometry, and re-shoot a short set at end of session to catch thermal drift (tripod_test_protocol.md 5).

6. **RUN:**

   ```
   scripts/calibrate_camera.py --images 'calib/*.png' --cols 9 --rows 6 --square 0.020 --out calib.json
   ```

   Every range and bearing the campaign produces is wrong until this passes. A uniform print-scale error is INVISIBLE in the RMS -- it is consistent with itself. The bars in step 2 and the measurement in step 4 are the only things standing between a 3% print error and a 3% error in every number.

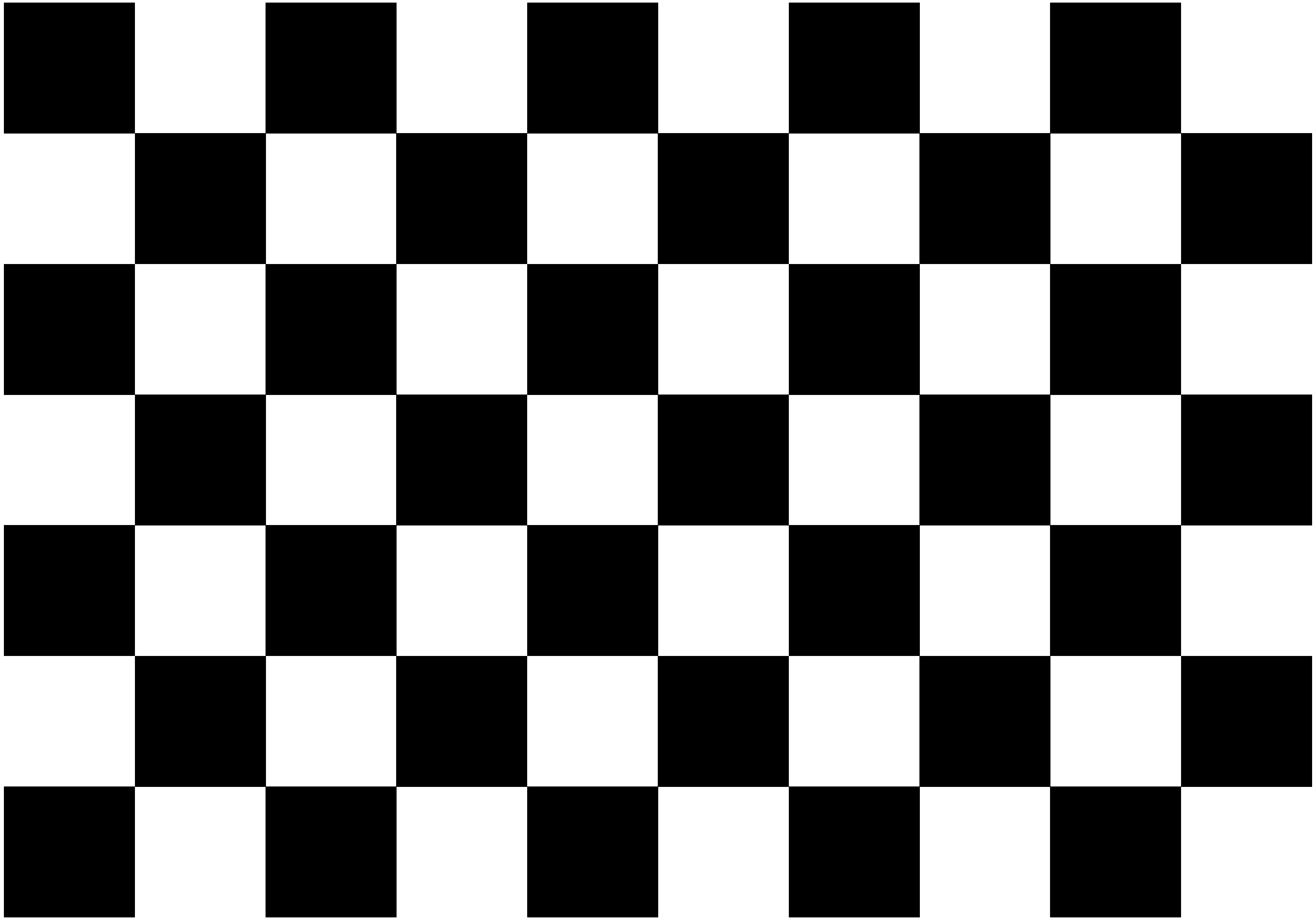
0 50 100 150

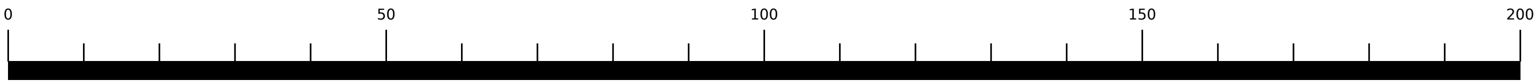
THIS BAR MUST MEASURE EXACTLY 150.0 mm

0 50 100 150 200

THIS BAR MUST MEASURE EXACTLY 200.0 mm

9 x 6 inner corners, 20.0 mm squares -- print at 100% / actual size





SCALE CHECK: exactly 200.0 mm